



UNIT 1

CHAPTER 1: Introduction to Data Structure

1.1 Data Structures Introduction

1.2 Classification of Data Structures

1.3 Operations of Data Structures

1.4 Algorithm

- 1.4.1 Algorithmic Notations
- 1.4.2 Control Structures
- 1.5 Complexity
- 1.5.1 Time-Space Tradeoffs

CHAPTER 2: Arrays

2.1 Definition of Arrays

2.2 Characteristics of Array Data Structure

2.3 Classification of Arrays

2.4 Representation of Linear Array in Memory

2.5 Operations on Linear Array

- Traversing
- Insertion
- Deletion
- 2.6 Searching Techniques
- Linear (Sequential) Search
- Binary Search
- Comparison of Searching Methods
- 2.7 Sorting
- Bubble Sort
- Selection Sort
- Insertion Sort
- 2.8 Merging Technique
- Merge Sort
- 2.9 Drawbacks & Advantages of Arrays
- 2.10 Two-Dimensional Arrays
- Memory Representation
- 2.11 Matrices
- Algebra of Matrices
- Sparse Matrices
- 2.12 Multi-Dimensional Arrays
- Memory Representation



UNIT 2

CHAPTER 3: Linked Lists

3.1 Linked List Definition

3.2 Comparison between Arrays and Linked List

- Drawbacks of Array / Advantages of Linked List
- Advantages of Array / Disadvantages of Linked List
- 3.3 Memory Representation of Linear Linked List
- 3.4 Types of Linked Lists
 - Singly Linked List
 - Header Linked List
 - Circular Linked List
 - Doubly Linked List
 - Two-Way Header List
- 3.5 Traversing Linked List
- 3.6 Free Storage List
- 3.7 Insertion in Linked List
 - Insert at Beginning
 - Insert after Given Node
 - Insert into Sorted Linked List
- 3.8 Deletion in Linked List
 - Delete First Node
 - Delete Last Node
 - Delete at Given Position
 - Delete by Given Item
- 3.9 Searching in Linked List
 - Unsorted List
 - Sorted List
- 3.10 Operations on Doubly Linked List
 - Insertion
 - Deletion
- 3.11 Advantages & Disadvantages of Linked List
- 3.12 Applications of Linked List as Polynomials
- 3.13 Algorithm for Addition of Polynomials

CHAPTER 4: Hashing and Collision

4.1 Hashing

- Concept of Hashing
 - Hashing Terminologies
 - 4.2 Hash Table
 - 4.3 Types of Hashing Functions
 - 4.4 Collision Resolution / Overflow Handling
 - Open Addressing (Linear Probing)
 - Chaining
 - Applications of Hashing
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UNIT 3

CHAPTER 5: Stacks

5.1 Stack Introduction & Definition

5.2 Representation of Stack

5.3 Array Representation

- PUSH Operation
- POP Operation
- PUSH & POP Algorithms
- Programming Implementation
 - 5.4 Linked List Representation
- PUSH
- POP
- Overflow Condition
- PUSH Algorithm
- POP Algorithm
 - 5.5 Overflow & Underflow
 - 5.6 Applications of Stack
 - 5.7 Stack Data Structure
 - 5.8 Polish Notation
 - 5.9 Reverse Polish Notation
 - 5.10 Infix, Prefix & Postfix Expressions
 - 5.11 Evaluation of Postfix Expression
 - 5.12 Conversion of Infix to Postfix Expression

CHAPTER 6: Recursion

6.1 Recursion

- Tail Recursion
- Indirect Recursion
- 6.2 Applications of Recursion
- 6.3 Runtime Stack
- 6.4 Tower of Hanoi
- 6.5 GCD (Euclid's Algorithm)
- 6.6 Recursive Common Divisor
- 6.7 Recursive Selection Sort
- 6.8 Advantages of Recursion
- 6.9 Disadvantages of Recursion



UNIT 4

CHAPTER 7: Queue

7.1 Queue

- Overflow & Underflow
- 7.2 Representation of Queue
- 7.3 Operations on Queue
- 7.4 Queue using Linked List
- 7.5 Circular Queue
- 7.6 Types of Queue
- Simple Queue
- Circular Queue
- Deque (Double Ended Queue)
- Priority Queue
- 7.7 Applications of Queue

CHAPTER 8: Graph

- 8.1 Introduction to Graph
- 8.2 Terminologies of Graph
- 8.3 Representation of Graph
- 8.4 Adjacency Matrix
- 8.5 Path Matrix
- 8.6 Warshall's Algorithm for Shortest Path
- 8.7 Linked Representation of Graph
- 8.8 Traversing a Graph

- Breadth First Search (BFS)

- Depth First Search (DFS)

CHAPTER 9: Trees

9.1 Trees

9.2 Terminologies Related to Binary Tree

9.3 Binary Tree

9.4 Types of Binary Tree

- Strictly Binary Tree
- Extended Binary Tree
- Complete Binary Tree
- Skewed Binary Tree

9.5 Binary Tree Representation

- Array Representation
- Linked Representation

9.6 Tree Traversal

- Inorder
- Preorder
- Postorder

9.7 Binary Search Tree (BST)

- Searching in BST
- Deletion in BST

9.8 Height Balanced Trees (AVL Trees)

- Representation
- Searching
- Insertion